

Git, Part 2: The Humans

User Study Day · Do Developers Really Know How to Use Git Commands? A Large-Scale Study Using Stack Overflow

A study of Git users

Open the class link (it ends in /thu) and type your first name.

Do Developers Really Know How to Use Git Commands? A Large-Scale Study Using Stack Overflow

Yang, Zhang, Pan, Xu, Zhou, Huang · ACM TOSEM 2022

Take the paper's survey

These are the paper's survey questions, which 92 developers answered.

Answer them for yourself on your laptop.

2. What is your professional area?
3. What is your highest degree?
4. How many years have you been using Git?
5. Which of the following levels do you consider your ability to use Git to be?
6. What are the main approaches you used to learn Git commands?
7. Do you have any suggestions for potential new developers on learning Git commands

Q1 of the form, the donation choice, is left out.

How the study was done

Collect	The authors collected 80,370 Stack Overflow questions about Git commands, asked between 2008 and 2020.
Count	A question counts for a command when the command's name appears in the question or in its accepted answer.
Measure	For each command they measured how often its questions were viewed, and how often a question had no accepted answer.
Survey	They also sent a survey to 508 developers, and 92 answered.

What the paper found

How many	Git questions have been a steady 0.4% of all Stack Overflow questions each year since 2010.
Who asks	Many askers have old Stack Overflow accounts. In 2020, 40% had joined more than 5 years earlier.
Most viewed	The most-viewed commands are mostly about undoing work: <code>revert</code> , <code>reflog</code> , <code>stash</code> , <code>clean</code> and <code>reset</code> .
Hardest	Rare commands most often lack an accepted answer. Among common commands, <code>credential</code> and <code>submodule</code> do.
How people learn	The 92 surveyed say they learned mostly on their own: 81.7% of the boxes they ticked were self-learning.

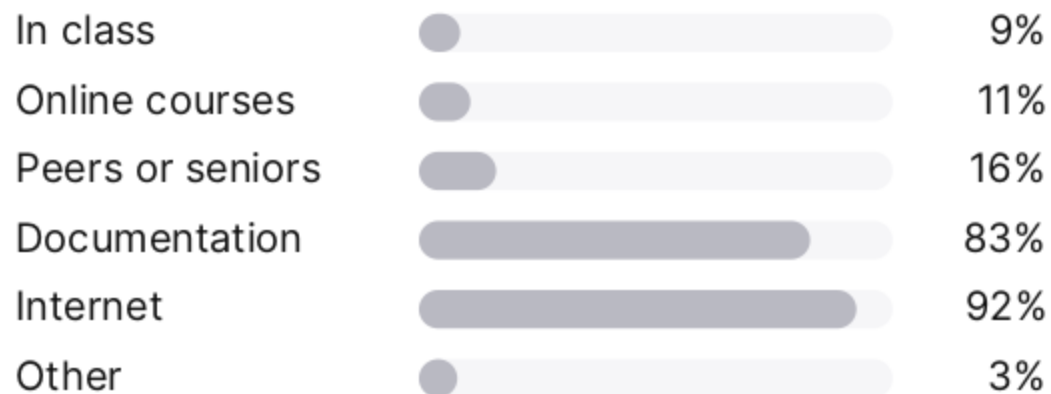
1 · THE STUDY

Your answers next to the paper's 92

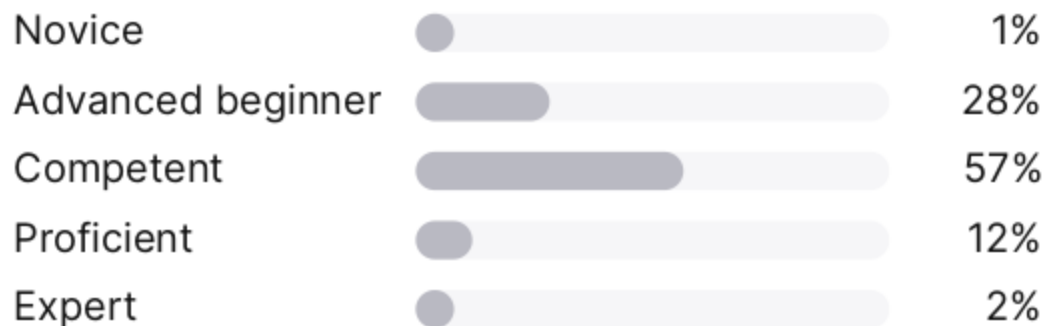
The paper's 92 were found mostly through Stack Overflow.

Their answers, like yours, are self-ratings and memories.

THE PAPER'S 92: HOW THEY LEARNED (SHARE OF PEOPLE)



THE PAPER'S 92: GIT LEVEL, SELF-RATED



Median years using Git: 8. In class, the app shows your answers next to these.

Would you trust your own answers?

Read 8 real posts

The paper counted each of these posts as a question about one command.

Read each post on your laptop and pick one of the three labels.

YES, STUCK ON IT

They were asking how to use this command.

NO. THEY NEEDED IT BUT DID NOT KNOW IT

The command would have fixed their problem, but they did not name it.

NO. IT IS NOT ABOUT IT

The command is only mentioned in passing.

On your laptop: 8 posts, each with the question, the accepted answer, and the command the paper counted it for.

Drawn at random from short, answered posts counted for the five most-viewed commands.

2 · CHECK IT YOURSELF

Your labels for the 8 posts

In 5 of the 8 posts, the command is named only in the accepted answer, not by the asker.

The paper still counts all 8 as questions about that command.

POST	COUNTED FOR	ASKER NAMES IT
git how to move back commit (6,163 views)	git reflog	Yes
How do I remove a single file from the staging area (undo git add)? (656,913 views)	git reset	Answer only
Undo a Git merge that hasn't been pushed yet (2,371,183 views)	git reflog	Answer only
How can i undo changes between two commits? (24 views)	git revert	Answer only
git 'merge' only existing files (1,378 views)	git clean	Yes
Git branch not working the way I expected it to (2,989 views)	git reset	Yes
revert a commit after push, checkout a commit behind and modification of files (46 views)	git revert	Answer only
Revert erroneous git checkout (45 views)	git stash	Answer only

Did the paper measure which commands are hard?

Sort 7 survey comments

The paper sorted the 65 comments from its survey into 6 categories.

On your laptop, put each of these 7 comments into one of those categories.

The paper's six categories:

Learning and improving from practice (18)

Making the best use of the internet (17)

Understanding basic concepts and principles first (14)

Emphasizing the importance of Git (9)

Mastering only the basic commands (5)

Having the basic needs satisfied by Git GUI tools (2)

2 · CHECK IT YOURSELF

Your sorting next to the paper's

The paper does not say how it sorted the comments, or whether two people agreed.

The usual practice is that two people sort independently, and the paper reports how often they agreed.

COMMENT	THE PAPER'S CATEGORY
#56 "Git is an extremely important tool to our routine as software ..."	Emphasizing the importance of Git
#32 "There are so many commands, I think it is better to learn them in ..."	Learning and improving from practice
#89 "Git is not a tool you have to go pretty deep to learn. Since it is a ..."	Mastering only the basic commands
#26 "The difficulty with git is that when problems arise, you have to dive ..."	Understanding basic concepts and principles first
#21 "All I want is to concentrate on my development, and dump the code ..."	Having the basic needs satisfied by Git GUI tools
#77 "Git is my VCS of choice, as I've learned how it works and have use ..."	Learning and improving from practice
#42 "Years on, I still constantly have to search the internet for help ..."	Making the best use of the internet

Would another team get the same categories?

BREAK

Break

We take five minutes, then work in groups.

Check one claim from the paper

Your group has one sentence from the paper, and the facts behind it.

Write what the facts really show, then what you would need to believe the claim.

RQ2, §3.2

"This suggests that even developers with years of development experience can have trouble using Git commands."

§1, FINDING 4; §3.4

"...for the more frequently-used commands, git credential and git submodule are among the most difficult ones."

ABSTRACT; §3.5

"Self-learning is the primary learning approach."

§3.5

"This result, although surprising, is consistent with the conclusion we obtained in RQ2, indicating that even experienced developers still have doubts about Git usage."

3 · JUDGE IT

The four claims, and what the data supports (1/2)

Nobody was watched using Git, so the data cannot answer the paper's title.

"This suggests that even developers with years of development experience can have trouble using Git commands."

THE DATA SUPPORTS

Some people with old Stack Overflow accounts asked for help. The claim is true as written.

"...for the more frequently-used commands, git credential and git submodule are among the most difficult ones."

THE DATA SUPPORTS

For git credential the share is above average (43.0% against 36.5% overall); for git submodule it is about average (37.5%).

Which claim holds up best?

3 · JUDGE IT

The four claims, and what the data supports (2/2)

Nobody was watched using Git, so the data cannot answer the paper's title.

"Self-learning is the primary learning approach."

THE DATA SUPPORTS

These 92 people say they learned mostly from the internet and the documentation.

"This result, although surprising, is consistent with the conclusion we obtained in RQ2, indicating that even experienced developers still have doubts about Git usage."

THE DATA SUPPORTS

Most of the 92 rate themselves competent or below (79 of 92).

Which claim holds up best?

Design a better undo

The most-viewed Git question asks how to undo the last commit.

Design one change, and say how you would test whether it helps.

YOUR CHANGE: WHAT WOULD THE USER SEE OR TYPE?

For example: a command that undoes the last thing Git did.

YOUR TEST: WHO WOULD TRY IT, DOING WHAT TASK?

For example: 20 students who just made a bad commit, half with the change.

WHAT WOULD YOU MEASURE, AND WHAT COULD FOOL YOU?

For example: time to recover; people get faster on the second try.

HOW WOULD YOU GET THE DATA: WATCH PEOPLE, LOG REAL USE, OR ASK THEM?

For example: screen recordings of the task.

Your designs

Each group reads its change, its test, its measure and risk, and its data. The app shows all of them.

Which of these could you test by watching people?

SUMMARY

What holds in the paper, and what does not

Holds

- ✓ There are many Git questions, and their share is steady.
- ✓ Undoing work is a real need.
- ✓ Long-time Stack Overflow users can still get stuck.
- ✓ The data is public, and we could check it ourselves.

Does not hold as written

- × The hardness ranking counts a command in the fix as the problem.
- × The most-viewed ranking rests on a few posts; one post lifts `reflog` from #8 to #2.
- × That developers learn on their own was asked of people found on Stack Overflow.
- × The comment categories have no stated method, and the “doubts” are self-ratings.
- × The title’s question stays open, because nobody was watched.

One idea to take home

On your laptop, finish this sentence in one line of your own.

When you build a tool for people, ...

THURSDAY

Thank you

What people ask shows where they get stuck, not what they can do.